Lec 2 2.3-5.4

概念自检

2.3 - 2.5

- function
- domain / codomain
  - image / preimage
    - $f(a) = b$
  - the image of S
    - $f(S) = f(s)|s \in S$
  - range
    - range 和 codomain 的区别是什么？哪个才对应中文里的“值域”？
- graph of function
- one-to-one / injective
  - 单射
  - $f(a) = f(b) \rightarrow a = b$
  - 到达域里的一个值只对应定义域里的一个值
- onto / surjective
  - 满射
  - 对于到达域里的每个值b，定义域中都存在至少一个a使得$f(a) = b$
  - Every b in B has a preimage
- one-to-one correspondence / bijection
  - 双射
  - |A| = |B|
- monotonic
  - monologue / monopoly
- inverse function
  - 只有双射存在逆函数
- composition of functions
  - $f \circ g$能被定义的必要条件：g的值域是f的定义域的子集
- floor function / ceiling function
- sequence
  - 可以留心一下定义严谨的表述
- geometric progression / arithmetic progression
  - 数列求和
- **cardinality**
  - 设 A=(a, b), B=(0, 1), 如何证明|A|=|B|？ 如果A = $(-\infty, +\infty)$ 呢?

- **countable**
  - aleph null $\aleph_0$ / countable infinte
  - 证明 $|Q^+| = |Z^+|$ **很重要！一定要会**
- properties of countable sets
- cantor diagonalization
  - $A = \{x | x \in (0,1) \wedge x \in R\}$, 求证 $|Z^+| < |A|$
    - 1. $|Z^+| \leq |A|$
      2. $|Z^+| \neq |A|$
  - 如何转化为直觉？$2^{\mathbb{N}}$与$\mathbb{N}$
  - 证明所有的 Java 程序是可数的 -> 证明存在不可计算的问题
- $\aleph_1$
- Schroder-bernstein theorem

> 证明：对于任意集合，其幂集的基比其本身的基更大

3.1 - 3.3

- algorithm
- big-O notation
  - $f(x) = O(g(x))$，哪一个是更加小一点的?
  - big-Omega
  - big-Theta
- computational complexity
- *P / NP / NP-complete （了解即可）

4.1 - 4.4

- division
  - factor / multiple
- divisor / dividend / quotient / remainder
  - divisor > 0, remainder ≥ 0
- division algorithm
- congruence relation
  - modulus
    - a mod m = b mod m
  - 同余运算的一些定理：
    - $a \equiv b \ (mod \ m) \leftrightarrow a = b + km$
    - If $a \equiv b \ (mod \ m)$ and $c \equiv d \ (mod \ m)$, then $a + c = b + d \ (mod \ m)$, $ac \equiv bd \ (mod \ m)$

- prime
- prime factoriazation
- greatest common divisor / least common multiple
- relatively prime
  - pairwise relatively prime
- Euclidean algorithm
  - $a = bq + r \rightarrow gcd(a, b) = gcd(b, r)$
- **Bezout's identity**
  - $gcd(a, b) = sa + tb$
  - Bezout coefficient
  - 如何解出s和t?
- linear congruence
  - $ax \equiv b \ (mod \ m)$, 其中 x 是变量
- **inverse**
  - An integer $\bar{a}$ such that $\bar{a}a \equiv 1 \ (mod \ m)$ is an *inverse of a modulo m*.
    - e.g. $5 \cdot 3 \equiv 1 \ (mod \ 7)$
  - If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Furthermore, this inverse is unique modulo m. 省流：互质 $\rightarrow$ 存在逆元
    - 逆元的求解思路是裴蜀定理的一种特殊情况
    - e.g.1 Find an inverse of 3 modulo 7
    - e.g.2 Find an inverse of 101 modulo 4620
    - 费马小定理
      - 若 p 是素数，且整数 a 不被 p 整除，则$a^{p-1} \equiv 1 \ (mod \ p)$
      - 标准形式：$a^p \equiv a \ (mod \ p)$，对任意整数 a 和素数 p 都成立
      - 与逆元求解的关系：$a \cdot a^{p-2} \equiv 1 (mod \ p)$
    - *欧拉定理，阶，primitive roots.....（了解即可）
  - Use inverse to solve congruences
    - e.g. solve $3x \equiv 4 \ (mod \ 7)$
  - **Chinese Remainder Theorem**
  - back substitution

5.1 - 5.4

- mathematical induction
  - first principle of mathematical induction
- well-ordering property
  - 良序性
- strong induction
  - second principle of mathematical induction / complete induction

    - using strong induction in computational geometry
- recursively defined function
- structral induction

习题整理

1. llf 25 quiz1

( ) (3) If $P(A) \in P(B)$, then $A \in B$. ($P(S)$ is the power set of S).

2.3 - 2.5

2. llf 24 quiz1

( )(7) For all positive real numbers x and y, $\lfloor x \cdot y \rfloor \leq \lfloor x \rfloor \cdot \lfloor y \rfloor$.

3. llf 24 quiz1

( ) (8)The function $f(n) = 3\lfloor n/3 \rfloor$ from Z to Z is a one-to-one function (injection), where Z is the set of integers.

4. zyw 25 期中

ii. Construct an injection $(0,1) \times (0,1) \rightarrow$ (0,1).(Hint: Elements of (0,1) can be represented as $0.a_1a_2\ldots$)(4')

5. zsy 25 quiz1

Let $x \in R$, $f(x) = [x]$, find:

- (1) $f^{-1}(\{1,2,3\})$
- (2) $f^{-1}(\{x \mid 1 < x < 2\})$

6. zsy 25 quiz1

Let $B = \{7k \mid k \in Z^+\}$, the set of positive integers that are multiples of seven. $A = Z^+ - B$, constructing a one-to-one correspondence function from A to Z^+.

- $A = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 22, 23, \cdots\}$
- $Z^+ = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, \cdots\}$

7. llf 25 quiz1

( ) (6) Let $F=\{f \mid f: N \rightarrow \{0,1\}\}$,Where N is the set of natural numbers, then F is uncountable..

8. 2019 秋冬 期末

( ) The set of all functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable infinite.

9. llf 25 quiz1

( ) (5) The set of real numbers that are solutions of quadratic equations $ax^2 + bx + c = 0$, where a, b, and c are integers, is uncountable.

10. 2018 秋冬 期末

3. **(12%)** Prove that the two sets of real numbers $(0, 1)$ and $[a, b) = \{x \in \mathbb{R} | a \leq x < b\}$ have the same cardinality(基数).

3.1 - 3.3

11. llf 24 quiz1

( )(9) $\dfrac{3n-8-4n^3}{2n-1}$ is $O(n^2)$..

12. llf 24 quiz1

( )(10)If $f(x)$ is $O(g(x))$, then $2^{f(x)}$ is $O(2^{g(x)})$

13. llf 24 quiz1

(5) Give a big-O estimate for the following function. For the function g in your estimate $f(x)$ is $O(g(x))$, use a simple function g of smallest order.

$$(n^3+n^2 \log n)(\log n+1) + (17 \log n+19)(n^3+2)$$

14. llf 25 quiz1

( ) (10) $f(n) = \sum_{j=0}^{n}(j^3 + j^2 + j + 1)$ is $O(n^4)$.

15. llf 25 quiz1

3. (4 points) **Prove that** $\dfrac{x^3+7x^2+3}{2x+1}$ **is** $\Theta(x^2)$.

4.1 - 4.4

16. llf 25 期中

Use Fermat's little theorem to find $9^{45} \ mod \ 23$.

17. llf 25 期中

Express gcd(84, 18) as a linear combination of 18 and 84.

18. zyw 25 期中

6 Show that if **a** and **b** are both positive integers, then $(2^a - 1) mod (2^b - 1) = 2^{a mod b} - 1$.(10')

19. zyw 25 期中

7 i. Compute 3^{2003} **mod** 7.(6')
ii. Compute the lowest digit (e.g.2 is the lowest digit of 8762; 3 is the lowest digit of 123) of $\lfloor 3^{2003} div 7 \rfloor$ in base-10.(4')

20. turing 25 quiz1

g)

$$2025^{2626} \equiv 1 \pmod{2027}$$

( )

21. turing 25 quiz1

6. (9%) Using the construction method from the Chinese Remainder Theorem proof, find all solutions to the system of congruences:

$$\begin{cases} x \equiv 1 \pmod 3, \\ x \equiv 2 \pmod 5, \\ x \equiv 3 \pmod 8. \end{cases}$$

5.1 - 5.4

22. llf 24 quiz1

(4) Find a recurrence relation with initial condition(s) satisfied by the following sequence $\{a_n\}$ (n = 1,2,3,...).

$a_n = n^2$

23. llf 25 期中

( ) The following is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers.

$$f(0) = 0,\ f(1) = 1,\ f(n) = 2f(n-1) \text{ for } n \geq 1$$

24. llf 25 期中

( ) Mathematical induction, strong induction, and well-ordering are three equivalent principles.

25. llf 25 期中

Let S be the subset of the set of ordered pairs of integers defined recursively by

Basic step:$(0,0) \in S$.

Recursive step:If $(a,b) \in S$,then $(a+2,b+3) \in S$ and $(a+3,b+2) \in S$.

Use structural induction to show that $5|a+b$ when $(a,b) \in S$.

26. turing 25 期中

8. (9%) Prove that every positive integer greater than 2 can be expressed as a sum of distinct Fibonacci numbers.

(从$F_1 = 1, F_2 = 2$开始)

27. 证明：$F_0F_1 + F_1F_2 + \cdots + F_{2n-1}F_{2n} = F_{2n}^2$

附录

1. 不止一位同学问过的作业题

4. Determine whether each of these sets is countable or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.

a) integers not divisible by 3

b) integers divisible by 5 but not by 7

c) the real numbers with decimal representations consisting of all 1s

d) the real numbers with decimal representations of all 1s or 9s

4. **a)** This set is countable. The integers in the set are ± 1, ± 2, ± 4, ± 5, ± 7, and so on. We can list these numbers in the order $1, -1, 2, -2, 4, -4, 5, -5, 7, -7, \ldots$, thereby establishing the desired correspondence. In other words, the correspondence is given by $1 \leftrightarrow 1$, $2 \leftrightarrow -1$, $3 \leftrightarrow 2$, $4 \leftrightarrow -2$, $5 \leftrightarrow 4$, and so on.

b) This is similar to part **(a)**; we can simply list the elements of the set in order of increasing absolute value, listing each positive term before its corresponding negative: $5, -5, 10, -10, 15, -15, 20, -20, 25, -25, 30, -30, 40, -40, 45, -45, 50, -50, \ldots$.

c) This set is countable but a little tricky. We can arrange the numbers in a 2-dimensional table as follows:

$.\overline{1}$	$.1$	$.11$	$.111$	$.1111$	$.11111$	$.111111$ $\ldots$
$1.\overline{1}$	1	1.1	1.11	1.111	1.1111	1.11111 $\ldots$
$11.\overline{1}$	11	11.1	11.11	11.111	11.1111	11.11111 $\ldots$
$111.\overline{1}$	111	111.1	111.11	111.111	111.1111	111.11111 $\ldots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

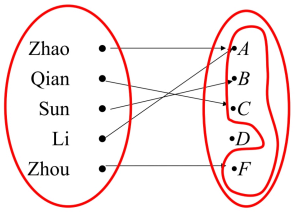
Thus we have shown that our set is the countable union of countable sets (each of the countable sets is one row of this table). Therefore by Exercise 27, the entire set is countable. For an explicit correspondence with

the positive integers, we can zigzag along the positive-sloping diagonals as in Figure 3: $1 \leftrightarrow .\overline{1}$, $2 \leftrightarrow 1.\overline{1}$, $3 \leftrightarrow .1$, $4 \leftrightarrow .11$, $5 \leftrightarrow 1$, and so on.

d) This set is not countable. We can prove it by the same diagonalization argument as was used to prove that the set of all reals is uncountable in Example 5. All we need to do is choose $d_i = 1$ when $d_{ii} = 9$ and choose $d_i = 9$ when $d_{ii} = 1$ or d_{ii} is blank (if the decimal expansion is finite).

2. notable slides

【**Example 1**】 Suppose that each student in a class is assigned a letter grade from the set $\{A,B,C,D,F\}$. Let g be the function that assigns a grade to a student.



Domain
Codomain
Range

TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x\rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty}, kx^{k-1}, \lvert x\rvert < 1$	$\frac{1}{(1-x)^2}$

2.5 Cardinality of Sets

【**Example 3**】 **Let A be the set of real numbers between a and b ($a<b$), and B be the set of real numbers between 0 and 1. Show that $|A| = |B|$.**

Question 4

Proof:

Let f be a function from A to B.

$$\frac{x-a}{b-a} = \frac{y-0}{1-0}$$

$$y = f(x) = \frac{x-a}{b-a}$$

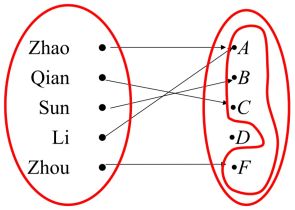
Then y is a bijection from (a, b) to $(0,1)$.

Hence, $|A| = |B|$.

The properties of the countable sets:

1) **No infinite set has a smaller cardinality than a countable set.**
2) **The union of two countable sets is countable.**
3) **The union of finite number of countable sets is countable.**
4) **The union of a countable number of countable sets is countable.**

〖**Example 1**〗Suppose that each student in a class is assigned a letter grade from the set $\{A,B,C,D,F\}$. Let g be the function that assigns a grade to a student.



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Codomain
Range

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〖**Example 5**〗 **Show that the set of finite strings S over a finite alphabet A is countably infinite.**

Proof:

Assume an alphabetical ordering of symbols in A. Show that the strings can be listed in a sequence. First list

1. **All the strings of length 1 in alphabetical order.**
2. **Then all the strings of length 2 in lexicographic (as in a dictionary) order.**
3. **Then all the strings of length 3 in lexicographic order.**
4. **And so on.**

This implies a bijection from N to S and hence it is a countably infinite set.

怎么证明任何集合的基数小于其幂集的基数?

引用 5 篇资料作为参考 >

以下用反证法证明任何集合的基数小于其幂集的基数：
设集合 A，其幂集记为 $\mathcal{P}(A)$（$\mathcal{P}(A)$ 是由 A 的所有子集构成的集合）。
步骤一：假设
假设集合 A 与它的幂集 $\mathcal{P}(A)$ 基数相等，那么根据基数相等的定义，存在一个一一对应（双射）$f: A \to \mathcal{P}(A)$，即对于 A 中的每一个元素 a，$f(a)$ 是 $\mathcal{P}(A)$ 中的一个元素（也就是 A 的一个子集），并且不同的 a 对应不同的子集，$\mathcal{P}(A)$ 中的每个子集也都有 A 中元素与之对应。
步骤二：构造特殊子集
定义集合 $B = \{x \in A | x \notin f(x)\}$。因为 B 是 A 的子集，所以 $B \in \mathcal{P}(A)$。
步骤三：推出矛盾
由于 f 是 A 到 $\mathcal{P}(A)$ 的双射，那么必然存在一个元素 $y \in A$，使得 $f(y) = B$。
现在考虑 y 与 B 的关系：

- 若 $y \in B$，根据 B 的定义 $\{x \in A | x \notin f(x)\}$，此时 y 应该满足 $y \notin f(y)$，但 $f(y) = B$，这就意味着 $y \notin B$，出现矛盾。
- 若 $y \notin B$，因为 $f(y) = B$，也就是 $y \notin f(y)$，按照 B 的定义 $\{x \in A | x \notin f(x)\}$，$y$ 又应该属于 B，同样出现矛盾。

所以假设不成立，即不存在集合 A 到其幂集 $\mathcal{P}(A)$ 的双射。又因为可以构造从 A 到 $\mathcal{P}(A)$ 的单射（例如，对 $a \in A$，令 $f(a) = \{a\}$，f 是 A 到 $\mathcal{P}(A)$ 的单射），所以集合 A 的基数小于其幂集 $\mathcal{P}(A)$ 的基数。这就是康托尔定理的结论。

分享

For example,

express gcd(252,198) = 18 as a linear combination of 252 and 198.

Solution:

First use the Euclidean algorithm to show gcd(252,198) = 18

I. 252 = 1·198 + 54
II. 198 = 3 ·54 + 36
III. 54 = 1 ·36 + 18
IV. 36 = 2 ·18

Now working backwards, from III and II above
18 = 54 − 1 ·36
36 = 198 − 3 ·54

Substituting the 2nd equation into the 1st yields:
18 = 54 − 1 ·(198 − 3 ·54)= 4 ·54 − 1 ·198

Substituting 54 = 252 − 1 ·198 (from I)) yields:
18 = 4 ·(252 − 1 ·198) − 1 ·198 = 4 ·252 − 5 ·198

裴蜀定理

Example: Find an inverse of 3 modulo 7.
Solution:

Because gcd(3,7) = 1, by Theorem 1, an inverse of 3 modulo 7 exists.

1. Using the Euclidian algorithm: 7 = 2·3 + 1.
2. From this equation, we get −2·3 + 1·7 = 1, and see that −2 and 1 are Bézout coefficients of 3 and 7.
3. Hence, −2 is an inverse of 3 modulo 7.
4. Also every integer congruent to −2 modulo 7 is an inverse of 3 modulo 7, i.e., 5, −9, 12, etc.

$\bar{a} \cdot a_1 \equiv 1 \pmod{m}$

求逆元1

Example: Find an inverse of 101 modulo 4620.
Solution:
First use the Euclidian algorithm to show that gcd(101,4620) = 1.

4620 = 45·101 + 75
101 = 1·75 + 26
75 = 2·26 + 23
26 = 1·23 + 3
23 = 7·3 + 2
3 = 1·2 + 1
2 = 2·1

Since the last nonzero remainder is 1, gcd(101,4620) = 1

Working Backwards:

1 = 3 − 1·2
1 = 3 − 1·(23 − 7·3) = − 1 ·23 + 8·3
1 = −1·23 + 8·(26 − 1·23) = 8·26 − 9 ·23
1 = 8·26 − 9 ·(75 − 2·26)= 26·26 − 9 ·75
1 = 26·(101 − 1·75) − 9 ·75
= 26·101 − 35 ·75
1 = 26·101 − 35 ·(4620 − 45·101)
= − 35 ·4620 + 1601·101

Bézout coefficients :
− 35 and 1601

1601 is an inverse of 101 modulo 4620

求逆元2

The Chinese Remainder Theorem

There are certain things whose number is unknown. When divided by 3, the remainder is 2; when divided by 5, the remainder is 3; when divided by 7, the remainder is 2. What will be the number of things?

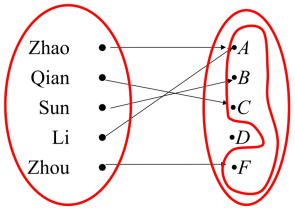


This puzzle can be translated into the solution of the system of congruences:

$x \equiv 2 \pmod 3$,
$x \equiv 3 \pmod 5$,
$x \equiv 2 \pmod 7$?

中国剩余定理

〖**Example 1**〗 Suppose that each student in a class is assigned a letter grade from the set $\{A,B,C,D,F\}$. Let g be the function that assigns a grade to a student.



Domain
Codomain
Range

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$\sum_{k=0}^{\infty} x^k, \lvert x\rvert < 1$	$\frac{1}{1-x}$
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The Chinese Remainder Theorem

Theorem 2: (*The Chinese Remainder Theorem*) Let $m_1,m_2,\ldots,m_n$ be pairwise relatively prime positive integers greater than one and $a_1,a_2,\ldots,a_n$ arbitrary integers. Then the system

$x \equiv a_1 \pmod{m_1}$
$x \equiv a_2 \pmod{m_2}$
.
.
.
$x \equiv a_n \pmod{m_n}$

has a unique solution modulo $m = m_1 m_2 \cdots m_n$.
(That is, there is a solution x with $0 \le x < m$ and all other solutions are congruent modulo m to this solution.)

Proof:
We'll show that a solution exists by describing a way to construct the solution.

The Chinese Remainder Theorem

first let $M_k = m/m_k$ for $k = 1,2,\ldots,n$ and $m = m_1 m_2 \cdots m_n$.

Since $\gcd(m_k, M_k) = 1$, there is an integer y_k, an inverse of M_k modulo m_k, such that

$$M_k y_k \equiv 1 \pmod{m_k}.$$

Form the sum

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_n M_n y_n.$$

Note that because $M_j \equiv 0 \pmod{m_k}$ whenever $j \neq k$, all terms except the kth term in this sum are congruent to 0 modulo m_k.
Because $M_k y_k \equiv 1 \pmod{m_k}$, we see that $x \equiv a_k M_k y_k \equiv a_k \pmod{m_k}$, for $k = 1,2,\ldots,n$.
Hence, x is a simultaneous solution to the n congruences.

$x \equiv a_1 \pmod{m_1}$
$x \equiv a_2 \pmod{m_2}$
.
.
$x \equiv a_n \pmod{m_n}$

The Chinese Remainder Theorem

Example: Consider the 3 congruences from Sun-Tsu's problem:

$x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$.

- Let m = 3· 5 · 7 = 105, M1 = m/3 = 35, M2 = m/5 = 21, M3 = m/7 = 15.
- We see that
  - 2 is an inverse of M1 = 35 modulo 3 since $35 \cdot 2 \equiv 2 \cdot 2 \equiv 1 \pmod 3$
  - 1 is an inverse of M2 = 21 modulo 5 since $21 \equiv 1 \pmod 5$
  - 1 is an inverse of M3 = 15 modulo 7 since $15 \equiv 1 \pmod 7$
- Hence,
  x = a1M1y1 + a2M2y2 + a3M3y3
  $= 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 \equiv 23 \pmod{105}$
- We have shown that 23 is the smallest positive integer that is a simultaneous solution. Check it!

Back Substitution

We can also solve systems of linear congruences with pairwise relatively prime moduli by rewriting a congruences as an equality, substituting the value for the variable into another congruence, and continuing the process until we have worked through all the congruences.

Example: Use the method of back substitution to find all integers x such that $x \equiv 1 \pmod 5$, $x \equiv 2 \pmod 6$, and $x \equiv 3 \pmod 7$.

Solution: the first congruence can be rewritten as $x = 5t + 1$, where t is an integer.

- Substituting into the second congruence yields $5t + 1 \equiv 2 \pmod 6$.
- Solving this tells us that $t \equiv 5 \pmod 6$.
- Using Theorem 4 again gives $t = 6u + 5$ where u is an integer.
- Substituting this back into $x = 5t + 1$, gives $x = 5(6u + 5) + 1 = 30u + 26$.
- Inserting this into the third equation gives $30u + 26 \equiv 3 \pmod 7$.
- Solving this congruence tells us that $u \equiv 6 \pmod 7$.
- By Theorem 4, $u = 7v + 6$, where v is an integer.
- Substituting this expression for u into $x = 30u + 26$, tells us that $x = 30(7v + 6) + 26 = 210u + 206$.

Translating this back into a congruence we find the solution $x \equiv 206 \pmod{210}$.